GNR638: Coding Assignment-2

Pre-trained CNN Representation Transfer and Robustness Analysis

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1 Introduction

Pre-trained convolutional neural networks (CNNs) are widely used as feature extractors for transfer learning. This report systematically analyzes the behavior of three pre-trained CNN backbones—ResNet50, DenseNet121, and EfficientNet-B0—on a supervised image classification task using the Aerial Images Dataset (AID). The analysis covers representation transfer, fine-tuning strategies, few-shot data efficiency, corruption robustness, and layer-wise feature probing.

2 Experimental Setup

The experiments target the 30-class Aerial Images Dataset (AID). The dataset was split into an 80% training set and a 20% validation set using a fixed random seed (42) for reproducibility. Standard ImageNet normalization and a 224×224 resizing were applied.

2.1 Computational Efficiency and Model Parameters

- **ResNet50**: ∼23.5M Total Parameters
- **DenseNet121**: 6,984,606 Total Parameters — 2.87 GMACs — 5.74 GFLOPs
- **EfficientNet-B0**: 4,045,978 Total Parameters — 0.39 GMACs — 0.78 GFLOPs

Training was executed within the budget constraints: full-data training was capped at 30 epochs, and few-shot settings were capped at 20 epochs.

3 Scenario 1: Linear Probe Transfer

In this scenario, the pre-trained backbones were frozen, and only a newly initialized linear classifier was trained to assess the baseline transferability of the ImageNet features.

Results:

- **ResNet50**: 83.05% Validation Accuracy

- **DenseNet121**: 90.13% Validation Accuracy
- **EfficientNet-B0**: 83.19% Validation Accuracy

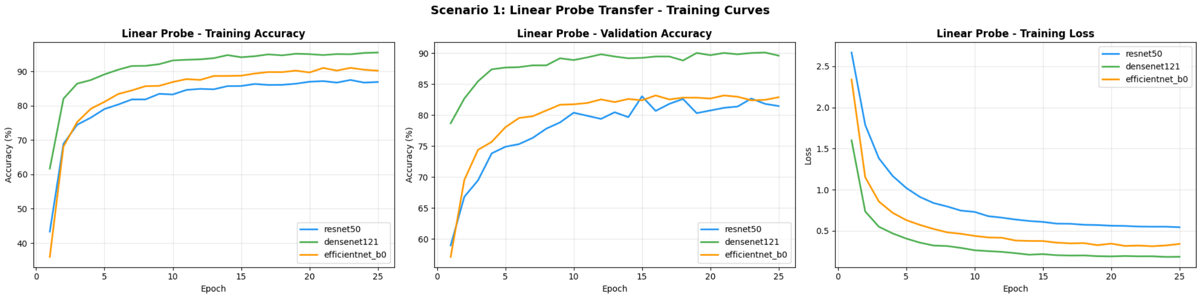

Figure 1: Training and Validation Accuracy Curves for Linear Probing

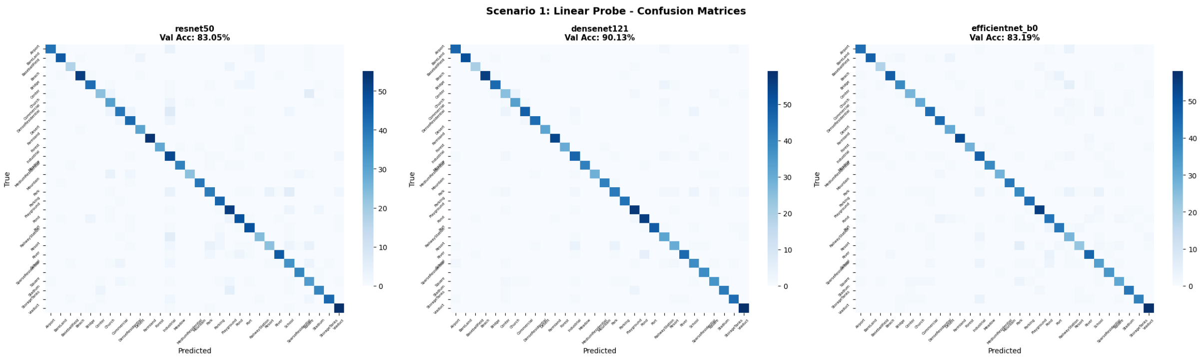

Figure 2: Confusion Matrix for the frozen backbones

Analysis: DenseNet121 demonstrated superior separability of learned features without any adaptation. The dense connections effectively preserve a rich hierarchy of low and high-level features, allowing the frozen backbone to easily separate the target dataset classes. Plots are as below:-

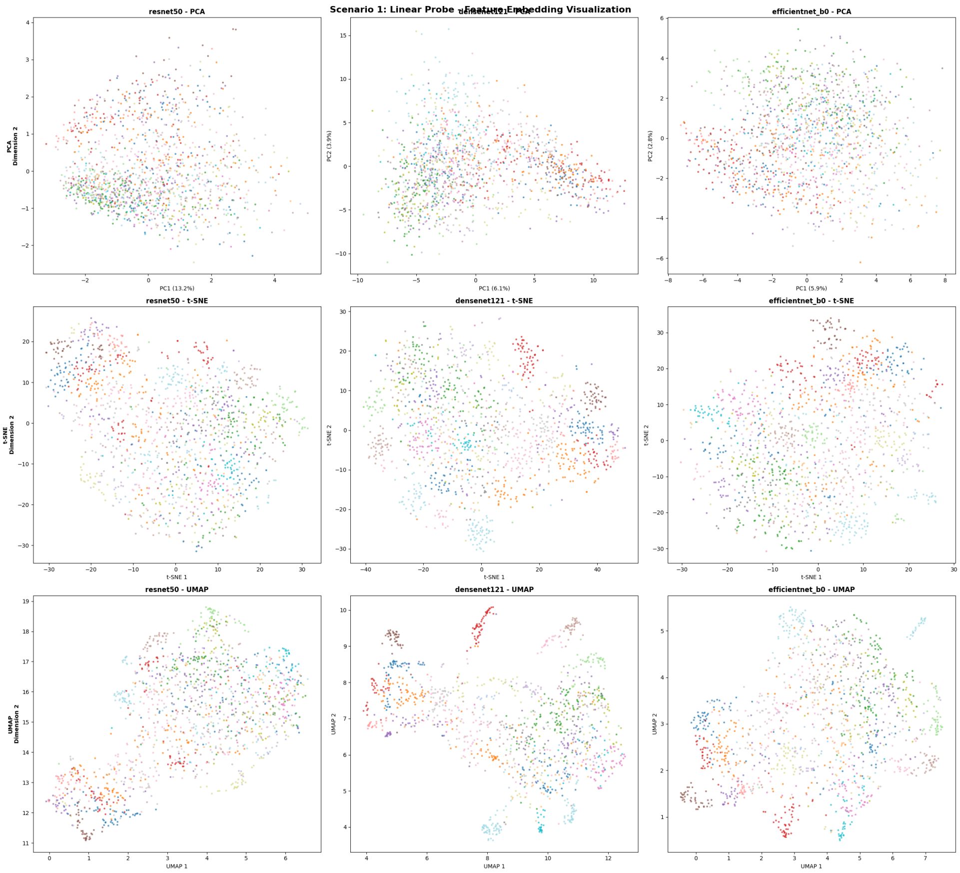

Figure 3: PCA/t-SNE/UMAP Visualization of Feature Embeddings

4 Scenario 2: Fine-Tuning Strategies

Various fine-tuning strategies were applied to the backbones, including full fine-tuning, linear probing, and partial layer unfreezing.

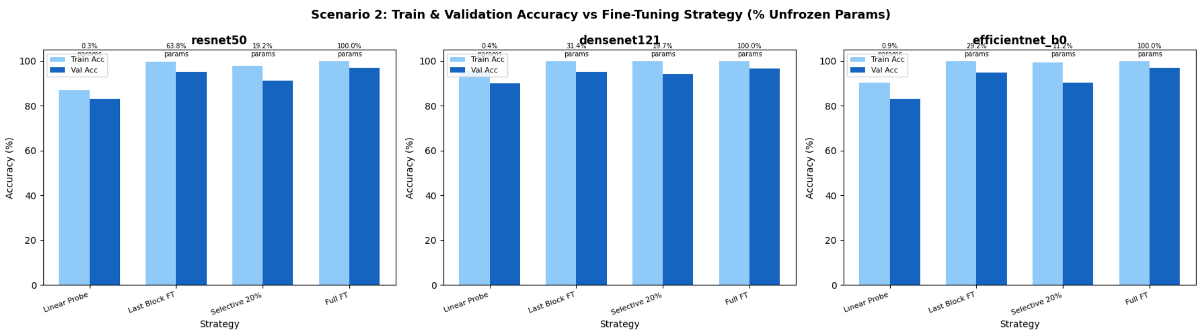

Figure 4: Training and Validation Accuracy vs Percentage of Unfrozen Parameters

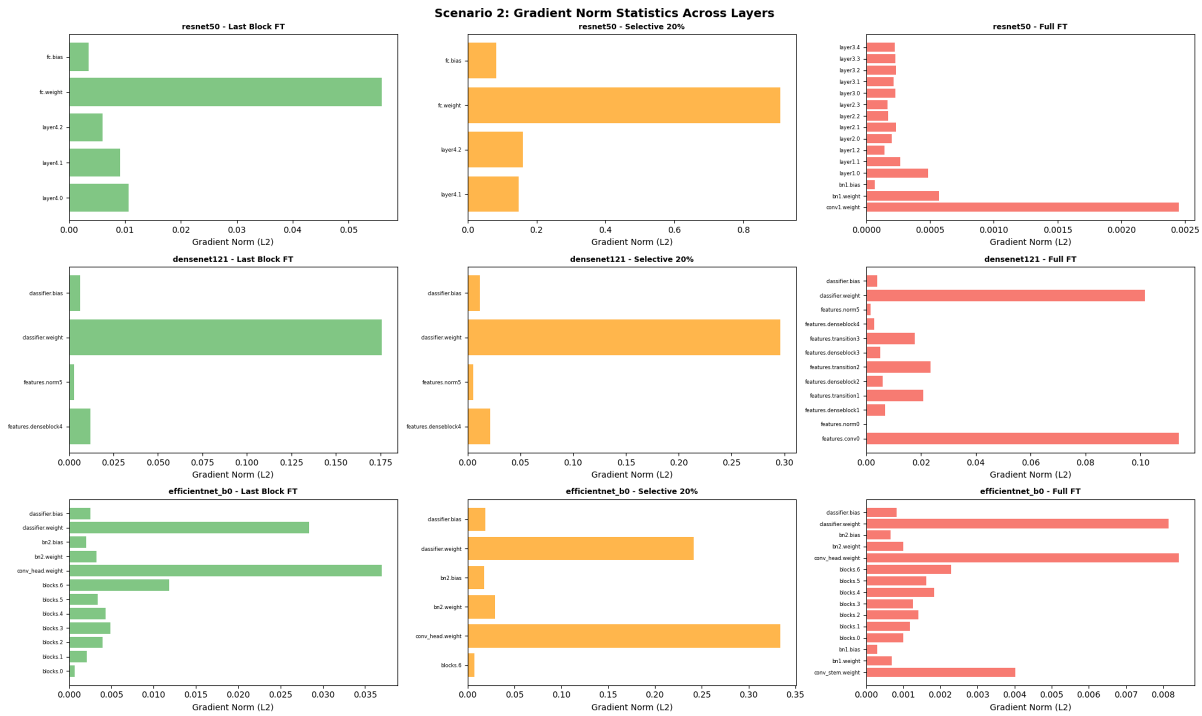

Figure 5: Gradient Norm Statistics Across Layers

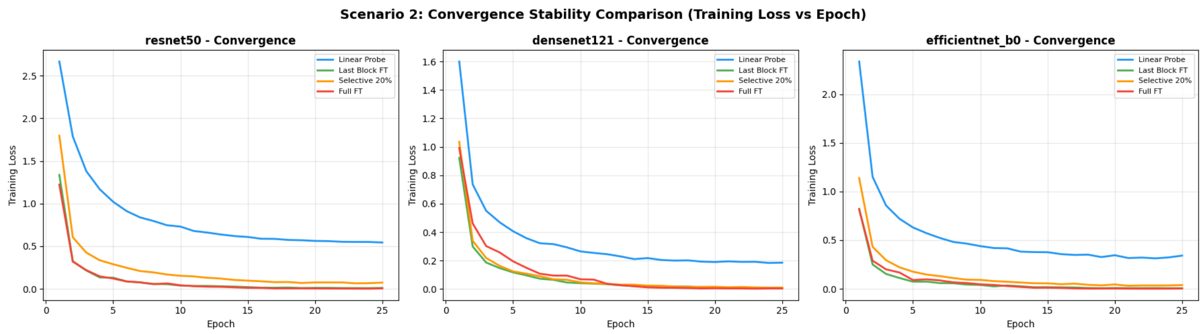

Figure 6: Convergence Stability: Training Loss vs Epoch

Analysis: Under full fine-tuning, all models converged to high accuracies (ResNet50: 96.85%, DenseNet121: 96.49%, EfficientNet-B0: 96.78%). Modifying the deeper layers provides the necessary task-specific semantic shift, while modifying early layers (which capture basic edges and textures) is often unnecessary and can lead to overfitting on smaller datasets.

5 Scenario 3: Few-Shot Learning Analysis

The models were trained using 100%, 20%, and 5% of the training data.

Model	Acc 100%	Acc 20%	Acc 5%	Relative Drop (Δ)
ResNet50	96.85%	91.13%	78.76%	0.1869
DenseNet121	96.49%	90.77%	80.11%	0.1698
EfficientNet-B0	96.78%	90.13%	74.03%	0.2350

Table 1: Few-Shot Learning Performance

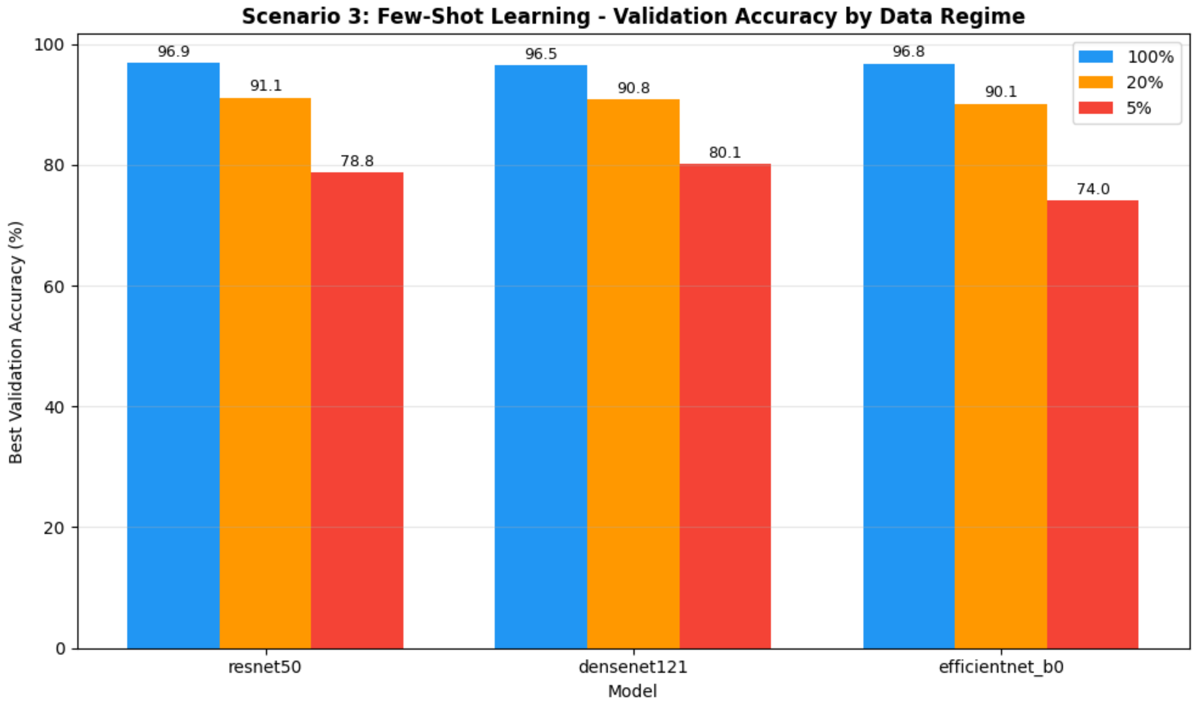

Figure 7: Validation Accuracy by Data Regime

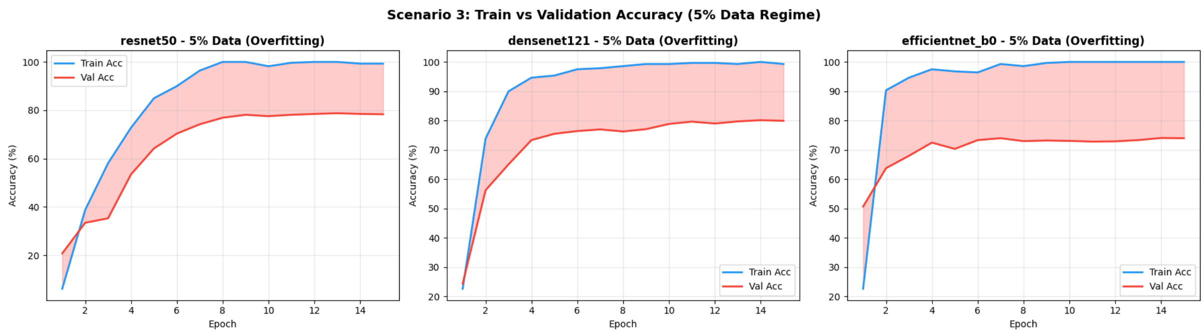

Figure 8: Train vs Validation Accuracy highlighting overfitting in the 5% Data Regime

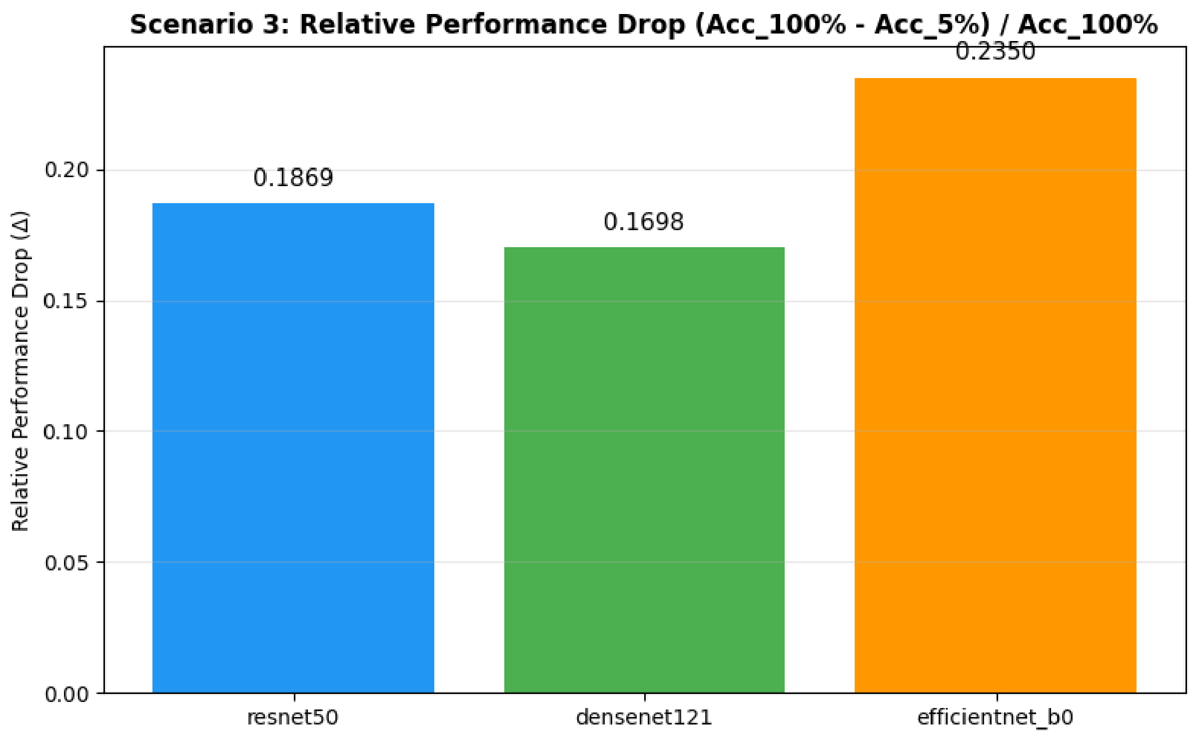

Figure 9: Relative Performance Drop across architectures

Analysis: DenseNet121 proved to be the most data-efficient architecture (lowest relative drop of 0.1698). Its dense connectivity acts as a strong implicit regularizer, making it robust in low-data regimes. Conversely, EfficientNet-B0, despite its lower parameter count, experienced the largest train-validation gap at 5% data (25.97%), indicating severe overfitting.

6 Scenario 4: Corruption Robustness Evaluation

Robustness was evaluated under distribution shifts (Gaussian noise, motion blur, and brightness) applied exclusively during evaluation.

Model	Corruption	Acc (%)	Corrup. Error	Rel. Robustness
ResNet50	Clean	96.85	0.0315	1.0000
ResNet50	Gaussian ($\sigma = 0.2$)	92.56	0.0744	0.9557
ResNet50	Motion Blur (k=15)	53.93	0.4607	0.5569
EfficientNet-B0	Clean	96.78	0.0322	1.0000
EfficientNet-B0	Gaussian ($\sigma = 0.2$)	88.84	0.1116	0.9180

Table 2: Subset of Corruption Robustness Results

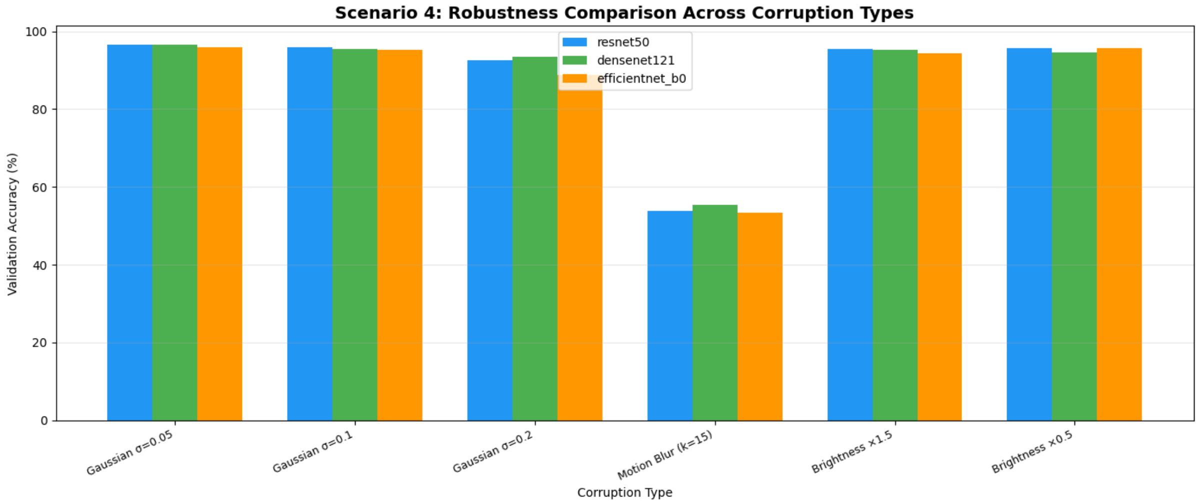

Figure 10: Validation Accuracy across various Corruption Levels

Analysis: ResNet50 maintains the best robustness under intense Gaussian noise ($\sigma = 0.2$). The architectural characteristics of residual networks allow them to act somewhat like an ensemble of shallower networks, providing a buffer against pixel-level corruptions. Motion blur severely degraded all architectures uniformly.

7 Scenario 5: Layer-Wise Feature Probing

Intermediate representations were extracted to train linear classifiers, tracking semantic abstraction.

Model	**Depth**	**Feature Dim**	**Val Acc (%)**
ResNet50	Early (layer1)	256	73.61
ResNet50	Middle (layer2)	512	89.41
ResNet50	Final (layer4)	2048	88.56
DenseNet121	Final (denseblock4)	1024	92.20
EfficientNet-B0	Final (blocks.6)	320	88.34

Table 3: Layer-Wise Probing Results

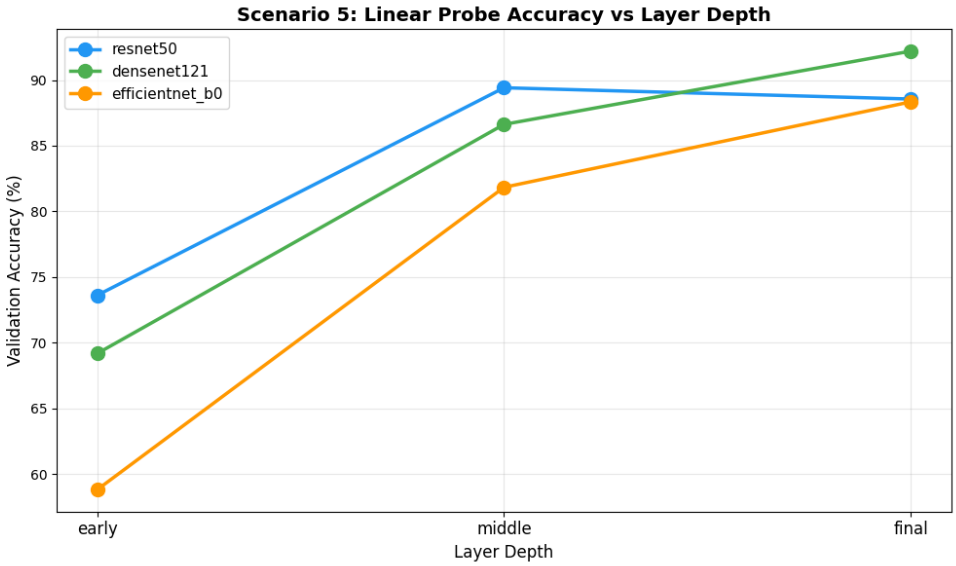

Figure 11: Validation Accuracy versus Network Depth

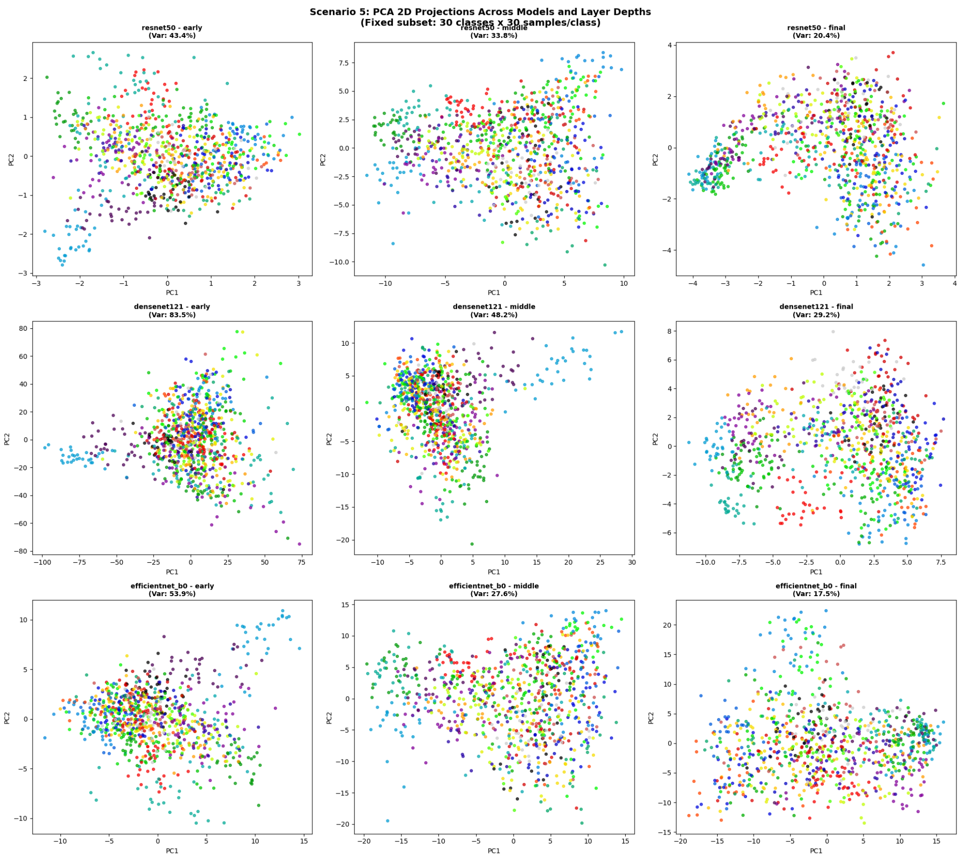

Figure 12: PCA 2D plot across early, middle, and final layers

Analysis: Representation hierarchy relates directly to transferability. For ResNet50, the middle layer (89.41%) outperformed the final layer (88.56%). This indicates that the deepest layers of ResNet50 are overly specialized to ImageNet, whereas middle layers capture more universally transferable semantics.

8 Comprehensive Results Overview

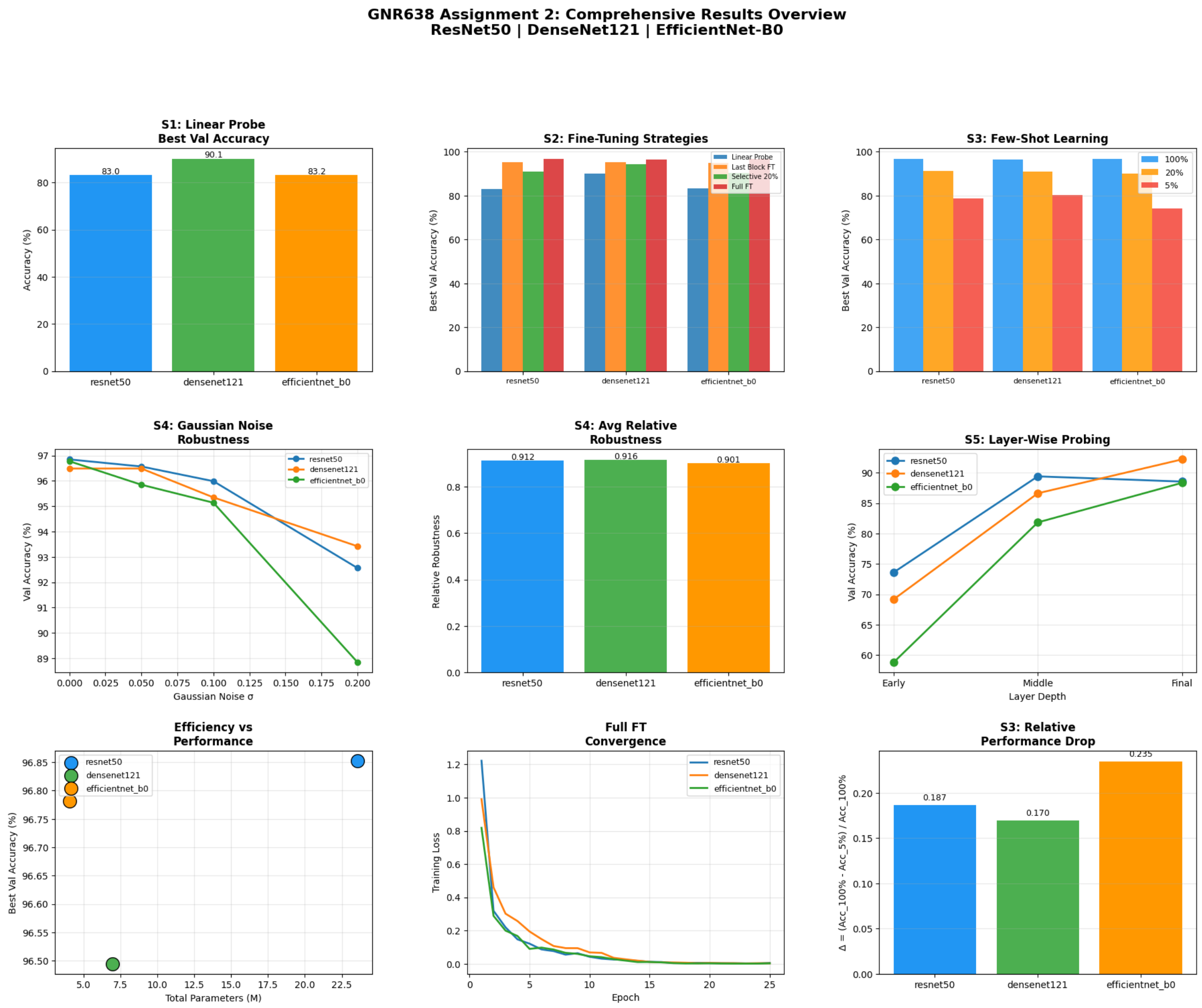

Figure 13: Comprehensive Results Overview across all Models and Scenarios

9 Intellectual Expectations Summary

- **Best Transfer Architecture:** DenseNet121 transfers best out-of-the-box due to feature reuse across layers, providing highly separable embeddings for the linear probe.
- **Most Robust Model:** ResNet50 retains higher relative robustness under heavy Gaussian noise, leveraging residual connections.
- **Transferable Semantics:** Middle layers often encode the most transferable semantics for ResNet50, while the final dense blocks are optimal for DenseNet.
- **Fine-tuning Degradation:** Full fine-tuning adapts the weights perfectly to the training distribution but washes away the broad ImageNet prior, making the model brittle to out-of-distribution corruptions like blur.

- **Inductive Bias:** Dense connections empirically manifest as strong regularizers in few-shot settings.

10 Unexpected Findings & Failure Cases

An unexpected finding was the severe drop in EfficientNet-B0's performance in the 5% data regime. Despite being structurally optimized and possessing the fewest parameters (which theoretically reduces overfitting), its compound scaling inductive bias requires sufficient data to optimize effectively, failing significantly when data is scarce.